10. Inserting a new node in a BST

Problem Statement

Write Recursive Approach to Insert New Node in a Binary Search Tree.

Given a binary search tree (BST) and a value to be inserted, write a recursive algorithm to insert a new node with the given value into the BST while maintaining its properties.

Examples

1. BST Before Insertion:



Input Node: 5 Output BST:



Explanation: The input node with value 5 is inserted as the left child of node 7.

2. BST Before Insertion:



```
  / \   \
 1   5   9
```

Input Node: 4 Output BST:

```
     6
    / \
   3   8
  / \   \
 1   5   9
    /
   4
```

Explanation: The input node with value 4 is inserted as the left child of node 5.

3. BST Before Insertion: Empty BST (null)

   Input Node: 2

   Output BST:

```
   2
```

   Explanation: The input node with value 2 becomes the root of the BST.

Try it yourself

Try solving this question here:

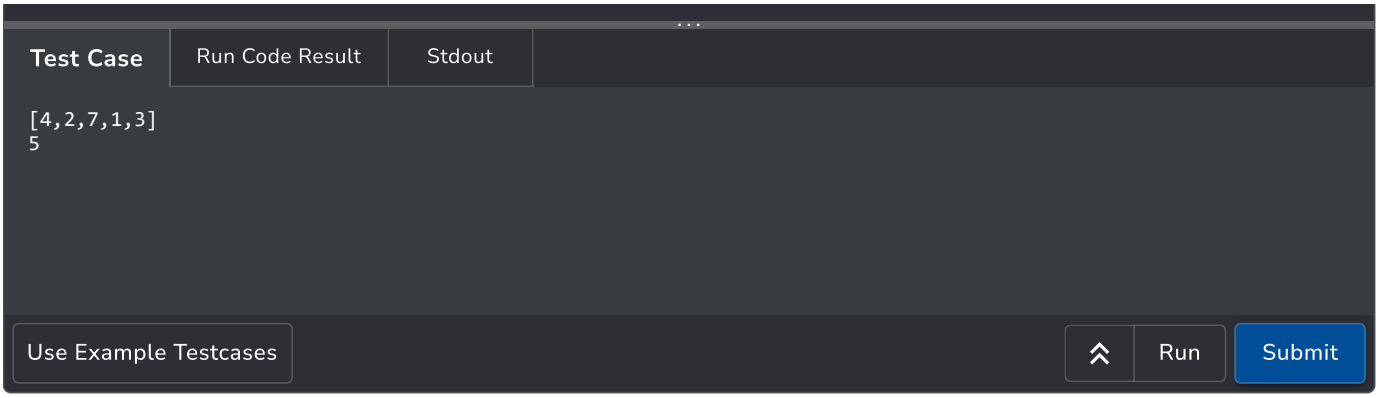
Java

```java
/*class TreeNode {
    int val;
    TreeNode left;
    TreeNode right;

    TreeNode(int val) {
        this.val = val;
    }
}*/

public class Solution {
    public TreeNode insert(TreeNode root, int value) {
          // TODO: Write your code here
        return root;
    }
}

```



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Solution: Binary Search

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Solution: Inserting A New Node In A BST

Mark as Completed